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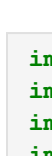
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📄 project-overview

This project aims to analyze a dataset of Aircraft accidents and incidents records to determine in which aircraft to invest in. This dataset was downloaded from Kaggle 📁 [dataset](#) and consists of 88859 rows and 31 columns.

To ensure data quality and reliability, we began by cleaning the dataset; this involved removing irrelevant columns and rows, handling missing values, and identifying and excluding outliers that could skew our analysis.

We conducted exploratory data analysis (EDA) using tools such as .head(), .value_counts(), and basic filtering techniques to better understand the structure and content of the data. We also used grouping and subsetting methods to uncover patterns and compare aircraft performance across different variables such as manufacturer, aircraft type, and year. Our final goal was to identify aircraft models that consistently showed low incident rates or minimal risk factors, based on multiple criteria derived from our analysis. These insights can help inform strategic decisions for stakeholders considering aircraft investments.



questions-of-interest

Before running my analysis, I have selected a range of questions that spike my interest and could help me in finding the aircraft less risky for investment.

- What are the aircraft categories in the dataset?
- How many accidents are there by aircraft?
- How likely is it for a 1-engine aircraft to be accidented?
- How likely is it for an aircraft to be destroyed in an accident?
- What is the death rate by accident for each aircraft category?

```
In [196...] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [197...] df = pd.read_csv("AviationData.csv",encoding = "ISO-8859-1",low_memory = False, parse_dates = True)
```

```
In [198...] df.head()
```

```
Out[198...]
```

	Investigation.Type	Accident.Number	Event.Date	Aircraft.damage	Aircraft.Category	Number.of.Engines	Engine.Type
0	Accident	SEA87LA080	1948-10-24	Destroyed	NaN	1.0	Reciprocating
1	Accident	LAX94LA336	1962-07-19	Destroyed	NaN	1.0	Reciprocating
2	Accident	NYC07LA005	1974-08-30	Destroyed	NaN	1.0	Reciprocating
3	Accident	LAX96LA321	1977-06-19	Destroyed	NaN	1.0	Reciprocating
4	Accident	CHI79FA064	1979-08-02	Destroyed	NaN	NaN	NaN

basic-cleaning

```
In [199...] #cleaning part
df.drop_duplicates(inplace = True)
#df.isna().sum() to check how many null values are there in each column.
df.dropna(inplace = True)
```

basic-filtering

```
In [200...] df["Years"] = df["Event.Date"].dt.year
df.drop(columns = "Event.Date").head(5)
```

```
Out[200...]
```

	Investigation.Type	Accident.Number	Aircraft.damage	Aircraft.Category	Number.of.Engines	Engine.Type
22	Accident	ATL82DA027	Substantial	Helicopter	1.0	Turbo Shaft
33	Accident	LAX82FA038	Destroyed	Airplane	2.0	Reciprocating
39	Accident	DCA82AA009	Destroyed	Airplane	2.0	Reciprocating
43	Accident	LAX82DA037	Substantial	Airplane	1.0	Reciprocating
44	Accident	ANC82FA012	Substantial	Airplane	1.0	Reciprocating

```
In [201...] TotalrecordsByAircraft = df["Aircraft.Category"].value_counts()
pd.DataFrame(TotalrecordsByAircraft)
```

```
Out[201...]
```

Aircraft.Category	
Airplane	8670
Helicopter	1514
Weight-Shift	60
Powered Parachute	36
Gyrocraft	34
Glider	28
Ultralight	6
Balloon	5
Rocket	1
Blimp	1

analysis

```
In [202...] accidentsByAircraft = df[df["Investigation.Type"]=="Accident"].loc[:,["Investigation.Type","Aircraft.Category"]]
accidentsByAircraft
```

```
Out[202...]
```

NumOfAcc		
Aircraft.Category	Investigation.Type	
Airplane	Accident	8402
Helicopter	Accident	1479
Weight-Shift	Accident	60
Powered Parachute	Accident	36
Gyrocraft	Accident	34
Glider	Accident	28
Ultralight	Accident	6
Balloon	Accident	5
Blimp	Accident	1
Rocket	Accident	1

some-remarks



On a subset of over 10355 records, 81.14% concerns airplane's accidents, 14.28% Helicopter's, 0.58% Weight-Shift's, 0.35 % Powered Parachute's , 3.3% Gyrocraft , and 0.27% Glider's. The others being irrelevant because of the small records they hold.

```
In [203...] TypeAccidentCount = df[df["Investigation.Type"]=="Accident"].loc[:,["Aircraft.Category","TypeAccidentCount"]]
```

```
Out[203...]
```

TypeOfAccCount		
Aircraft.Category	Aircraft.damage	
Airplane	Substantial	7638
Helicopter	Substantial	1317
Airplane	Destroyed	696
Helicopter	Destroyed	158
Airplane	Minor	66
Weight-Shift	Substantial	56
Powered Parachute	Substantial	32
Gyrocraft	Substantial	31
Glider	Substantial	27
Balloon	Substantial	5
Ultralight	Substantial	5
Helicopter	Minor	4
Gyrocraft	Destroyed	3
Powered Parachute	Destroyed	3
Weight-Shift	Destroyed	3
Airplane	Unknown	2
Glider	Destroyed	1
Blimp	Destroyed	1
Powered Parachute	Minor	1
Rocket	Destroyed	1
Ultralight	Destroyed	1
Weight-Shift	Minor	1

🔍 Airplane_Destroyed_Rate = **8.28%**

🔍 Helicopter_Destroyed_Rate = **10.68 %** 🤔

🔍 Weight-Shift_Destroyed_Damage_Rate = **5%**

🔍 Powered Parachute_Destroyed_Rate = **8.34%**

🔍 Gyrocraft_Destroyed_Rate = **8.82%**

🔍 Glider_Destroyed_Rate = **3.57%**

```
In [204...] #total fatal injuries by aircraft category and number of engines
Injuries = df[df["Investigation.Type"]=="Accident"].loc[:,["Number.of.Engines","Aircraft.Category","TotalFatalInjuries"]]
```

```
Out[204...]
```

AccidentCount		
Number.of.Engines	Aircraft.Category	
1.0	Airplane	7176
	Helicopter	1370
2.0	Airplane	1189
	Helicopter	109
1.0	Weight-Shift	60
	Powered Parachute	36
	Gyrocraft	34
	Glider	22
4.0	Airplane	22
3.0	Airplane	13
1.0	Ultralight	5
0.0	Glider	5
1.0	Balloon	3
8.0	Airplane	1
2.0	Ultralight	1
0.0	Airplane	1
2.0	Glider	1
	Balloon	1
1.0	Rocket	1
	Blimp	1
8.0	Balloon	1

🔍 On the 8402 Airplane accidents , most of them are from 1-Engine planes, and we can see(👀) that **0-Engine** and **8-Engine Airplane** only caused **1 accident each** !

🔍 We can Also see that across all the Aircrafts, **1-Engine type is the worst leading either most accidents or All** 😬!

```
In [205...] TotalFatalInjuries = df[df["Investigation.Type"]=="Accident"].loc[:,["Aircraft.Category","TotalFatalInjuries"]]
```

```
Out[205...]
```

TotalFatalInjuries	
Aircraft.Category	
Airplane	3137.0
Helicopter	578.0
Weight-Shift	28.0
Gyrocraft	9.0
Powered Parachute	6.0
Glider	4.0
Ultralight	2.0
Rocket	1.0
Balloon	0.0
Blimp	0.0

```
In [206...] TotalFatalInjuries["TotalFatalInjuries"].sum() == df["Total.Fatal.Injuries"].sum()
#This verifies that the sum of Injuries partitioned by each Aircraft adds up to all the injuries
```

```
Out[206...] True
```

📊 -Death_rate_by_accident = (Total.Fatal.Injuries/Total_Accidents)



- Airplane_Death_Rate_by_Accident = 37.34%
- Helicopter_Death_Rate_by_Accident = 39 %
- Weight-Shift_Death_Rate_by_Accident = 46.67% 🤔
- Powered Parachute_Death_Rate_by_Accident = 25%
- Gyrocraft_Death_Rate_by_Accident = 17%
- Glider_Death_Rate_by_Accident = 14.28%

graphs

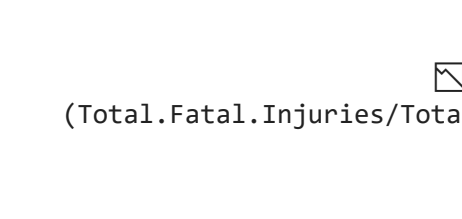
```
In [207...] done = df[["Years","Investigation.Type"]].groupby(["Years"]).count()
y = df["Investigation.Type"]
fig,ax = plt.subplots()
ax.plot(y, linestyle = "--", color = "r")
ax.set_title("Evolution of Accidents/Incidents over the years")
ax.set_ylabel("Accident/Incident counts")
ax.set_xlabel("Years (1982-2022)")
ax.set_facecolor("w")
```

```
# This gives a quick overview of the evolution of aircrafts failures (Accidents/incidents)
```



```
In [208...] fig, ax = plt.subplots()
ax.pie(df["Investigation.Type"].value_counts())
colors = ["Accidents", "Incident"]
plt.legend(colors)
plt.show()
```

```
#This shows the proportion of accidents against incidents of the aircrafts
```



```
In [209...] fig, axes = plt.subplots(figsize = (15,6))
axes.plot(df["Aircraft.Category"].value_counts()) # Thid graph shows the number of accidents/incidents by aircraft category
```

```
Out[209...] [<matplotlib.lines.Line2D at 0x22215ca75e0>]
```



results

I would suggest the company to start investment in Glider because it has proven to be the less risky throughout our Analysis. If we consider that each aircraft should have at least 25 records to be considered in the analysis, Glider holds the minimum number on (accidents + incidents) in our 10355 subset. Then Glider destroyed rate on an accident is 3.57% which is very low compared to the others, that means that if a Glider were the make an accident, the probability of the aircraft to be destroyed is 3.57%. Moreover, the sum of all fatal injuries caused by Gliders is the subset is only 4 ;, leaving us with a death rate of 14.28% , the lowest.

However one recommendation i would make to the busines is to avoid 1-engine gliders which account for 22 accidents over the 28 records, the company should invest in 2-engines gliders which have only caused one accident over the 28 recorded !

The business should also consider diversifying its portfolio of aircrafts in the future by going for its next best option : Gyrocraft.

```
In [211...] df.to_excel('cleaned_data.xlsx', index=False)
```